

Reporte de Análisis Físico

Validación MC y Asimetría UMD/SD

Configuración de la Simulación

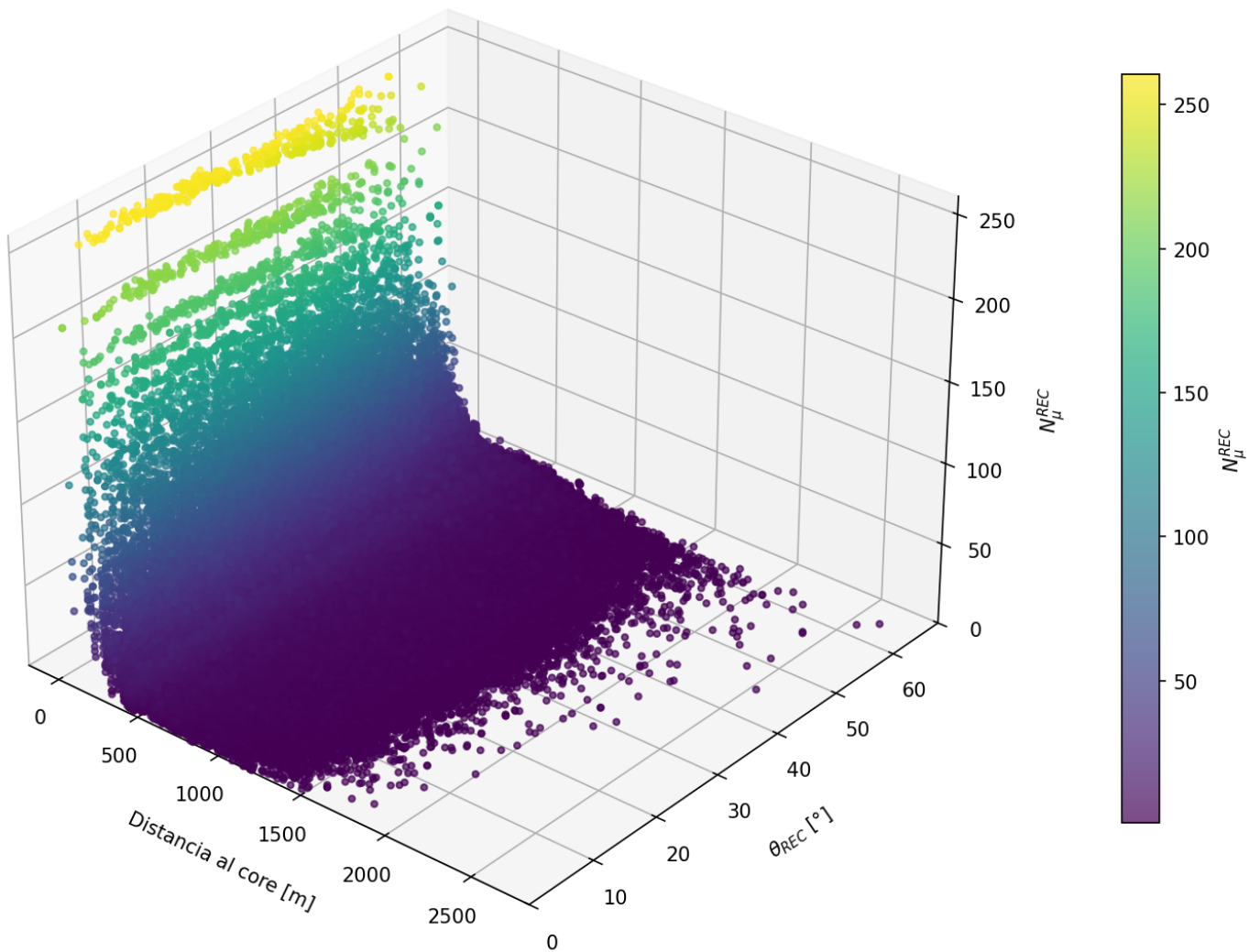
Parámetro	Valor
Modelo Hadrónico	QGSJetII-04
Primario	Helio
Rango Energía	17.5 - 18.0 log(eV)
Carpeta	
Eventos Totales	4,987
Módulos Totales	1,605,141

1. Limpieza de Datos

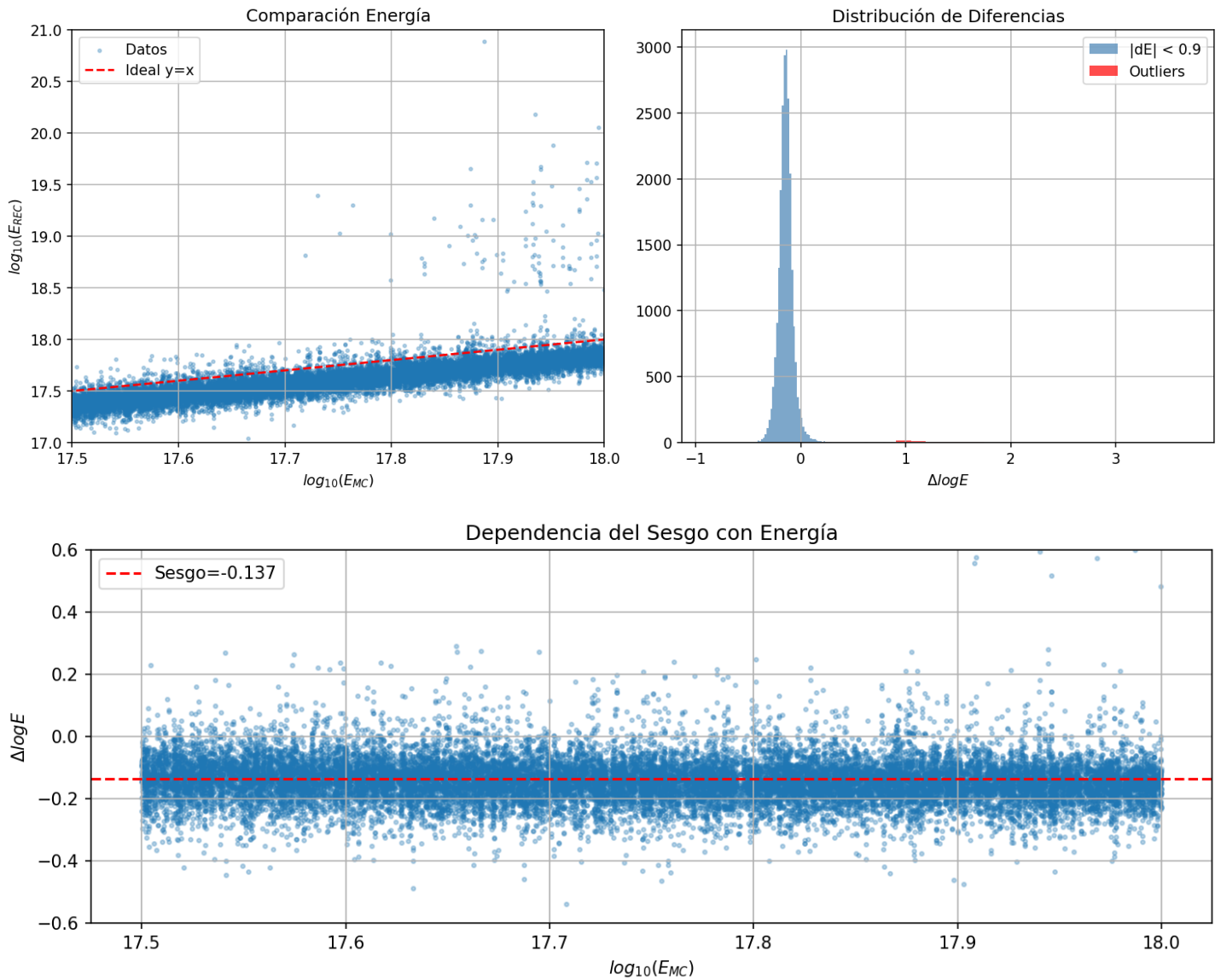
REPORTE DE LIMPIEZA

Métrica	Valor
Filas Iniciales	1605141
Filas Eliminadas	60519 (3.77%)
Filas Finales	1544622
Causa Principal	NaN en logE_REC (60519)

2. Visualización General

N_{μ}^{REC} vs distancia y θ 

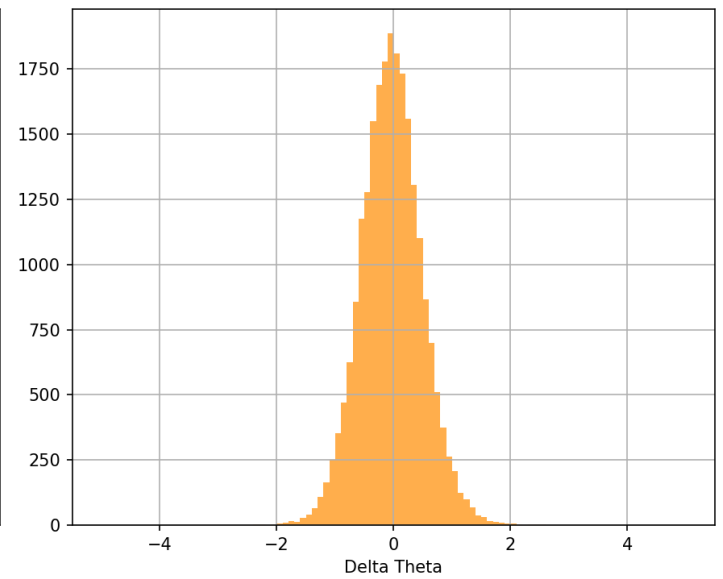
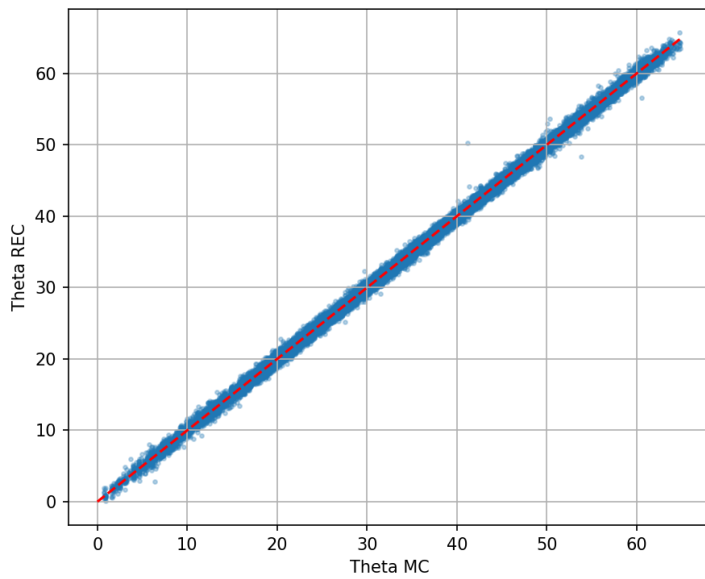
3. Resolución de Energía



Sesgo (Mean): -0.1367
 Resolución (Std): 0.1115
 Outliers (> 0.9): 57 (0.25%)

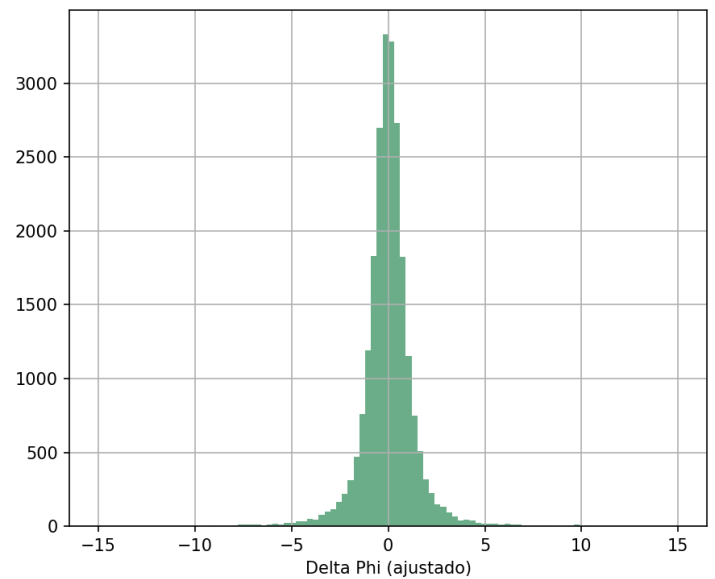
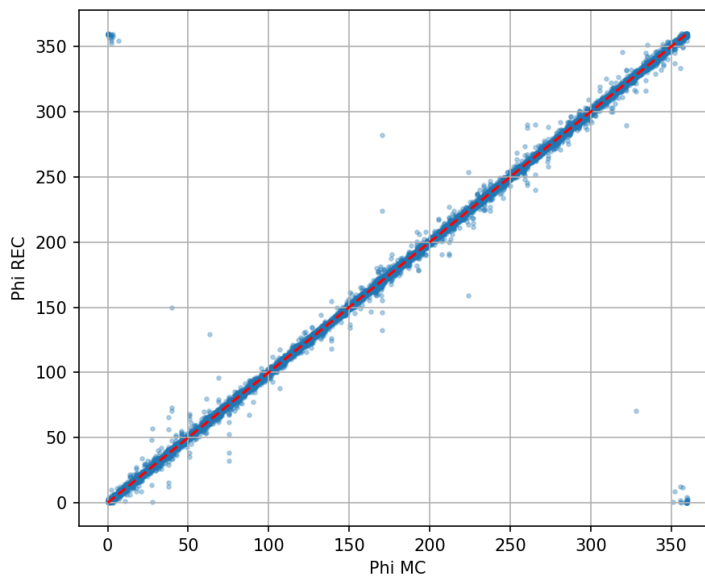
4. Resolución Angular

Theta (Cenital)



Sesgo: -0.0358 deg
Resolución: 0.5290 deg
Outliers ($> 10.0^\circ$): 0 (0.00%)

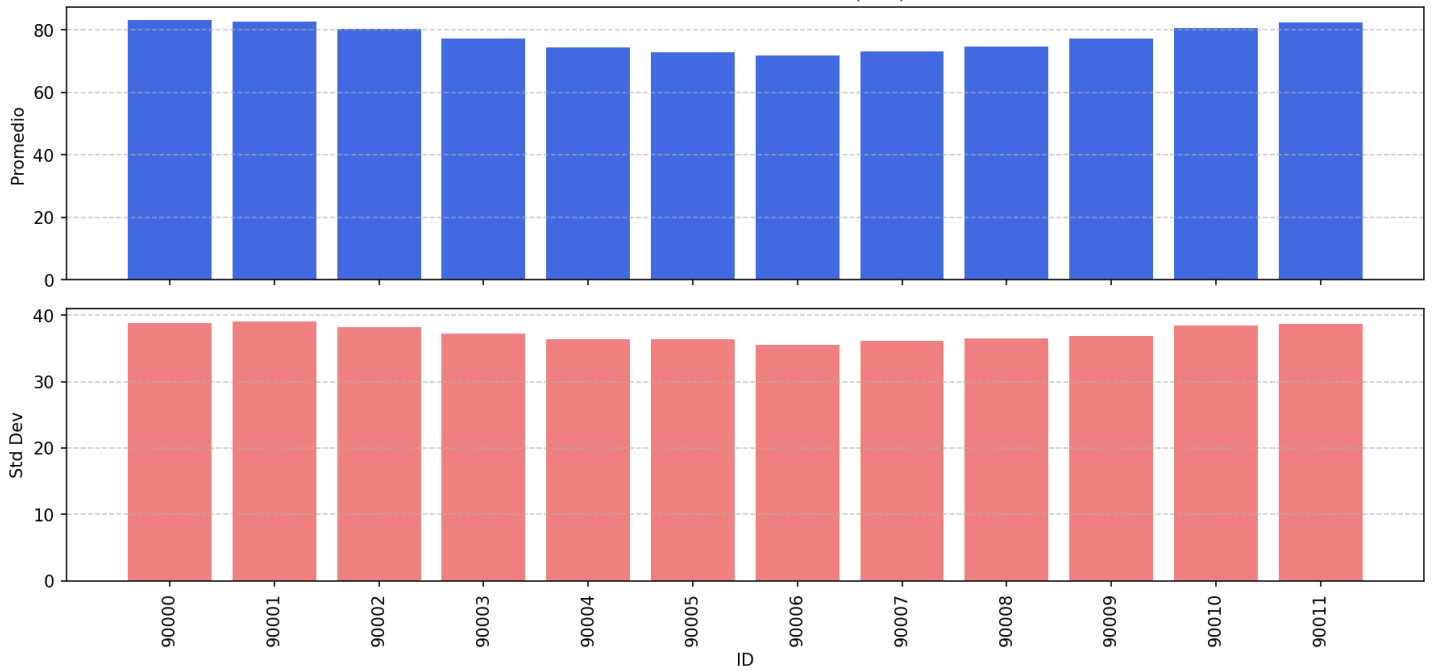
Phi (Azimutal)



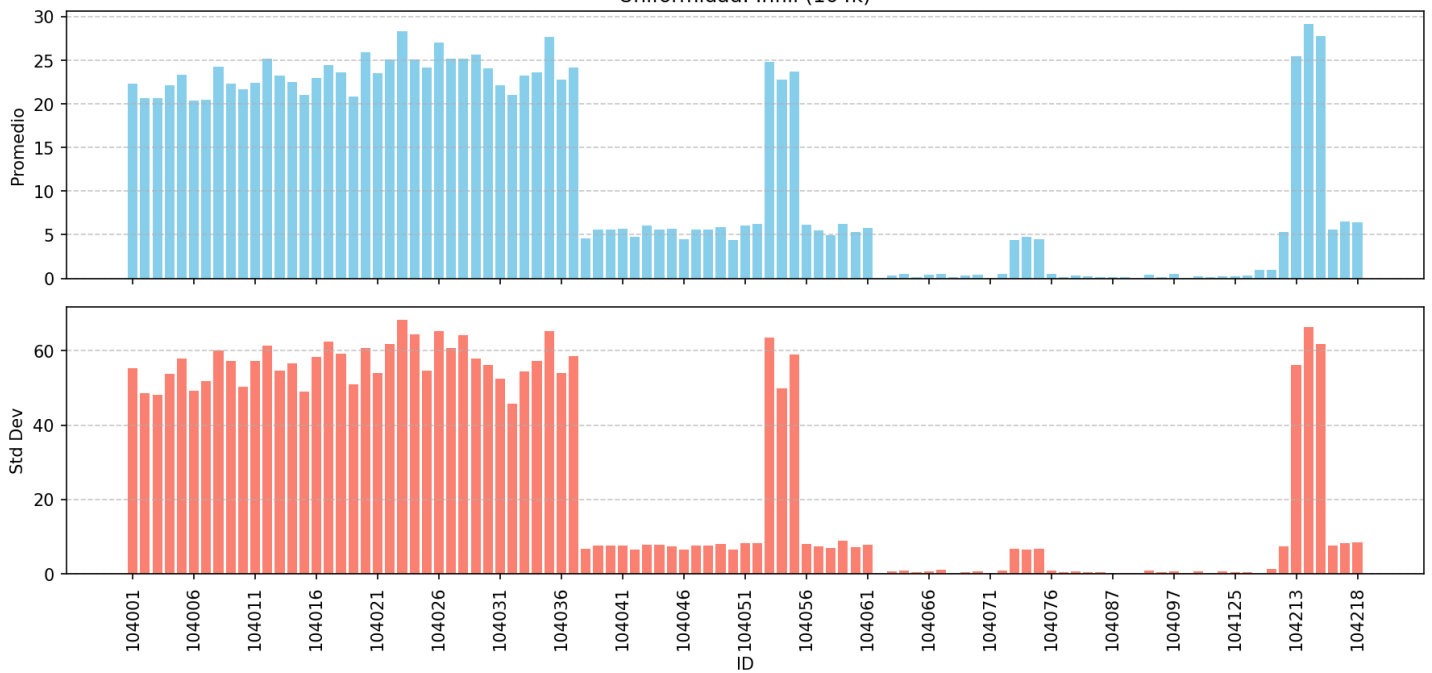
Sesgo (Median): -0.0012 deg
Resolución (MAD): 0.5729 deg
Outliers ($> 15.0^\circ$): 53 (0.23%)

5. Uniformidad por Counter

Uniformidad: Anillo Denso (90k)

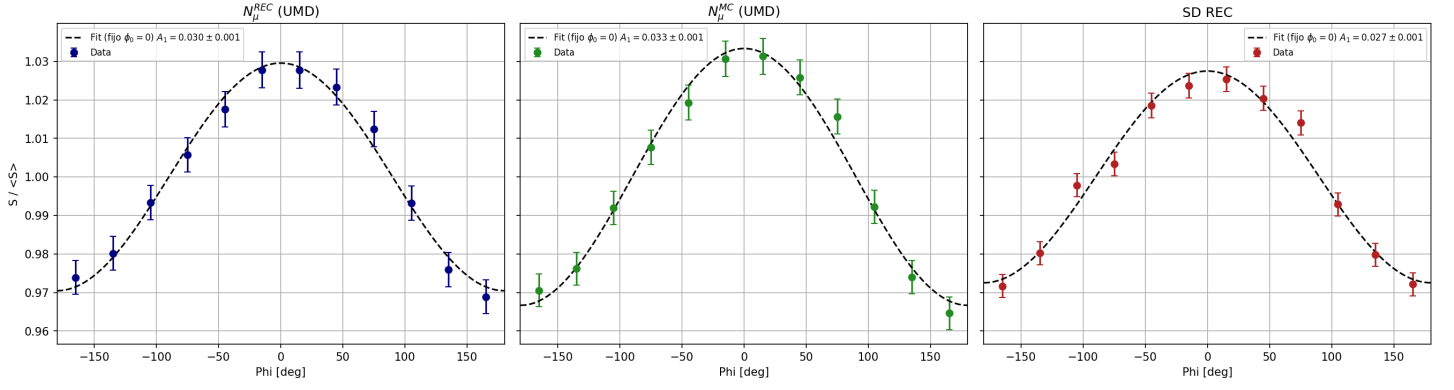


Uniformidad: Infill (104k)

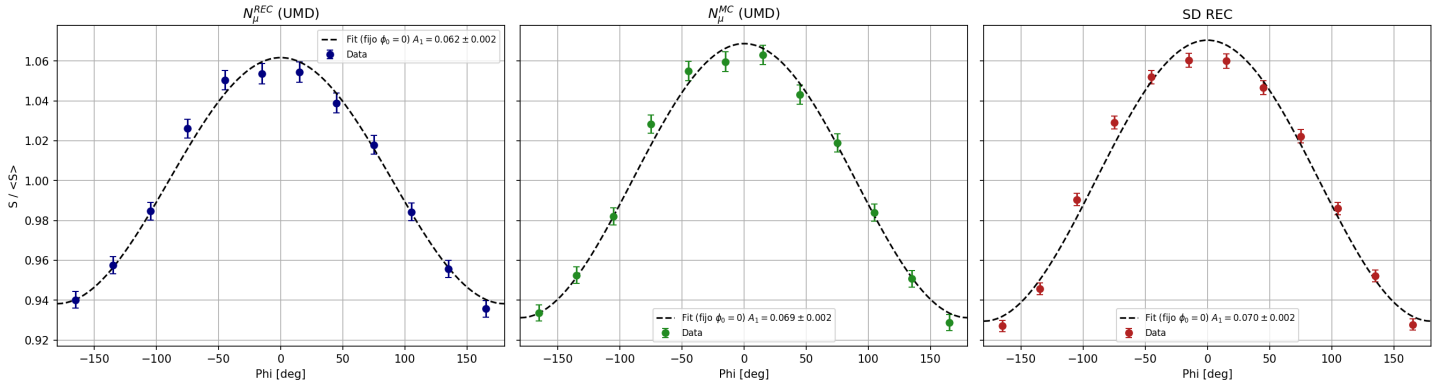


6. Asimetría Azimutal

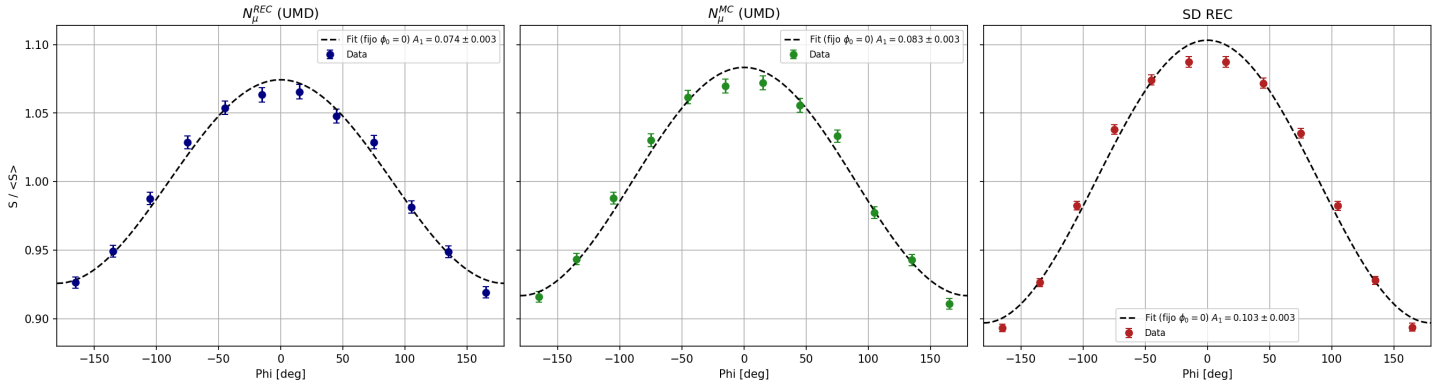
Bin 0: Theta 0.1 - 21.7 deg



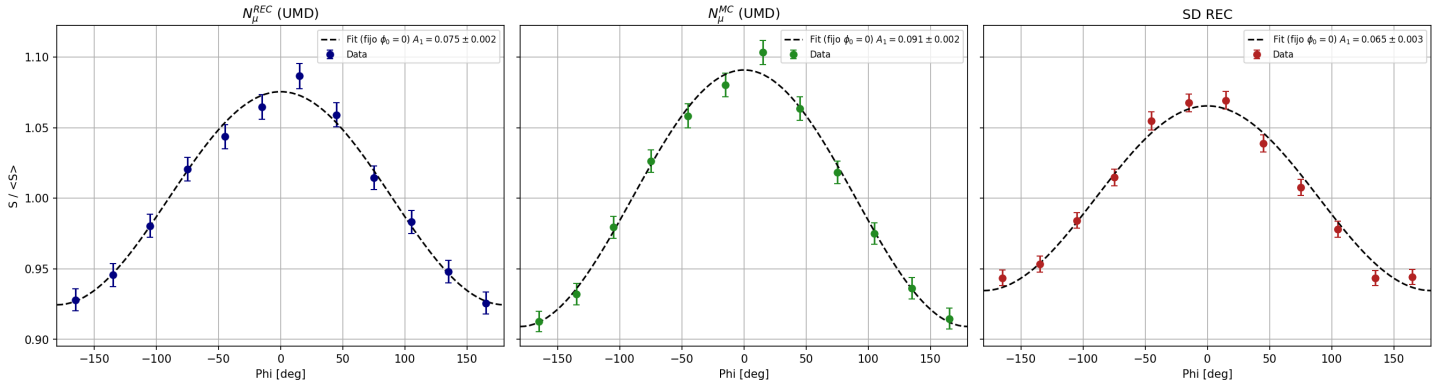
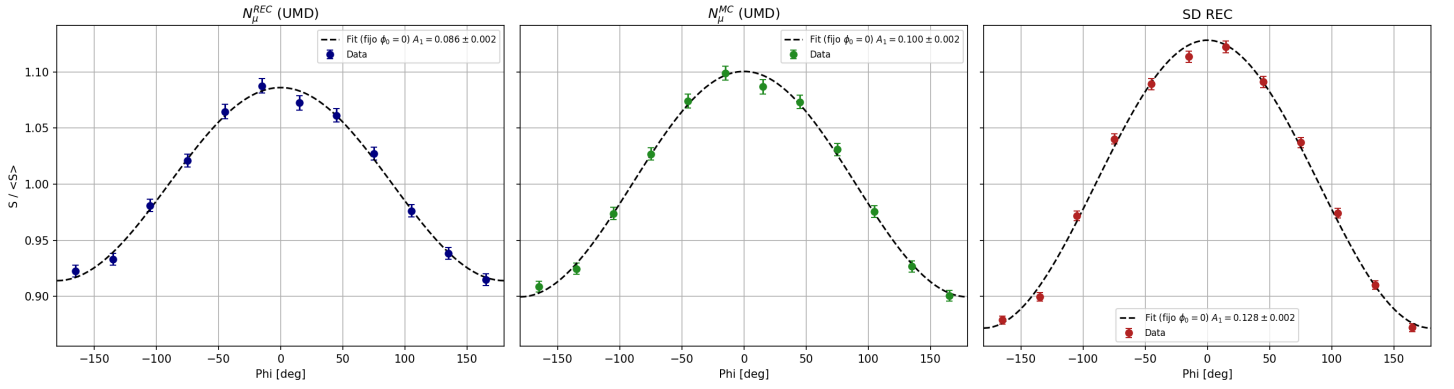
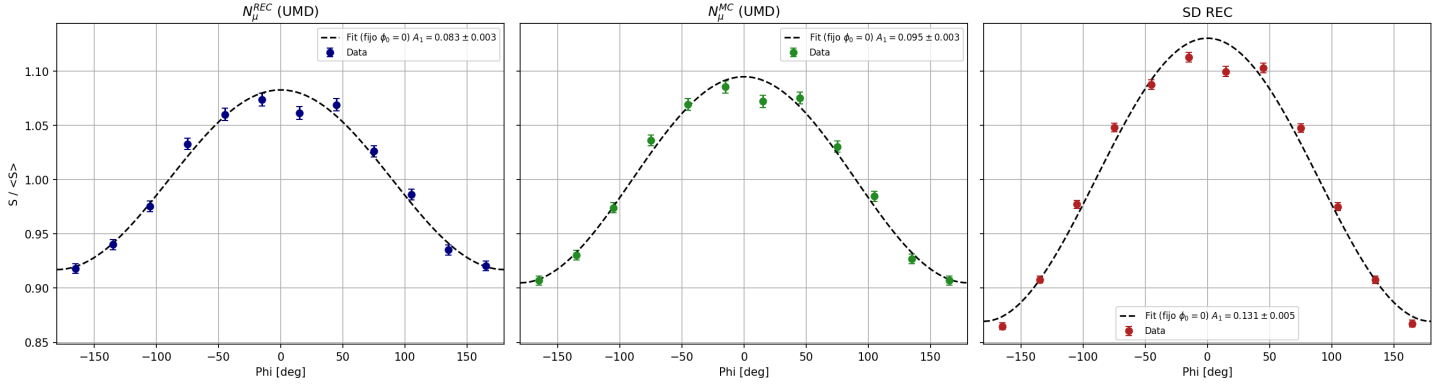
Bin 1: Theta 21.7 - 31.6 deg



Bin 2: Theta 31.6 - 39.9 deg



Bin 3: Theta 39.9 - 47.7 deg



7. Resumen de Ajustes (Comparación)

Valores de Amplitud A1

theta_range	A1_UMD_REC	A1_UMD_MC	A1_SD_REC
0.1-21.7	0.0296	0.0334	0.0275
21.7-31.6	0.0617	0.0688	0.0705
31.6-39.9	0.0743	0.0833	0.1032
39.9-47.7	0.0829	0.0951	0.1307
47.7-55.8	0.0860	0.1004	0.1282
55.8-65.0	0.0755	0.0909	0.0654

